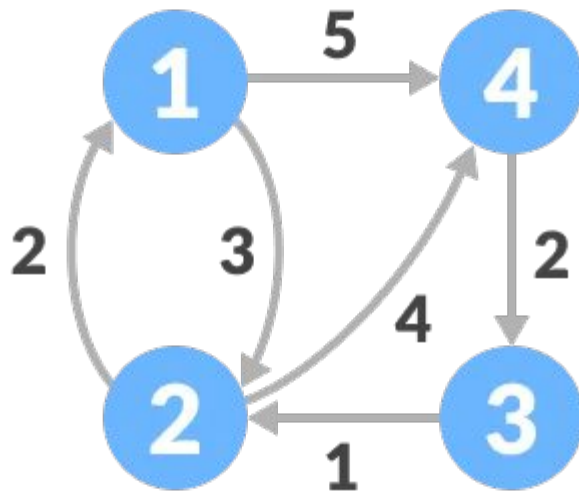
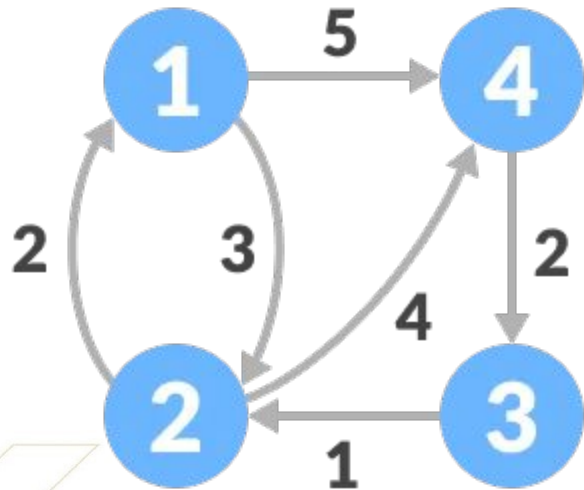


Floyd Warshall Algorithm



$$A^0 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 3 & \infty & 5 \\ 2 & 0 & \infty & 4 \\ \infty & 1 & 0 & \infty \\ \infty & \infty & 2 & 0 \end{bmatrix} \end{matrix}$$

Floyd Warshall Algorithm



$$A^k[i, j] = \min (A^{k-1}[i, j], A^{k-1}[i, k] + A^{k-1}[k, j])$$
$$A^1 [2,3]=\min(A^0 [2,3],A^0 [2,1]+A^0 [1,3])$$

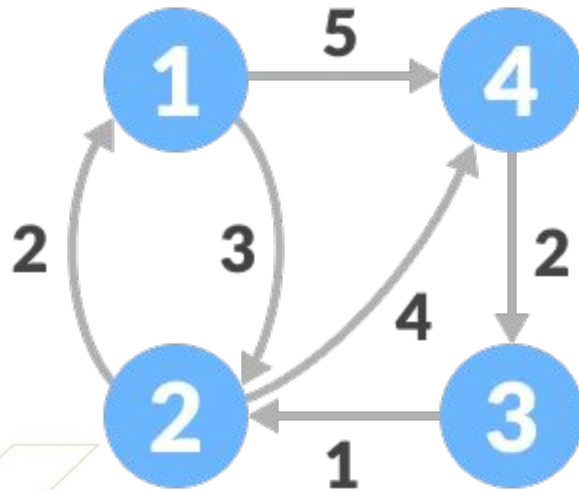
A^0

	1	2	3	4
1	0	3	INF	5
2	2	0	INF	4
3	INF	1	0	INF
4	INF	INF	2	0

A^1

	1	2	3	4
1	0	3	INF	5
2	2	0	INF	4
3	INF	1	0	INF
4	INF	INF	2	0

Floyd Warshall Algorithm



$$A^k[i, j] = \min (A^{k-1}[i, j], A^{k-1}[i, k] + A^{k-1}[k, j])$$
$$A^2 [1,4]=\min(A^1 [1,4],A^1 [1,2]+A^1 [2,4])$$

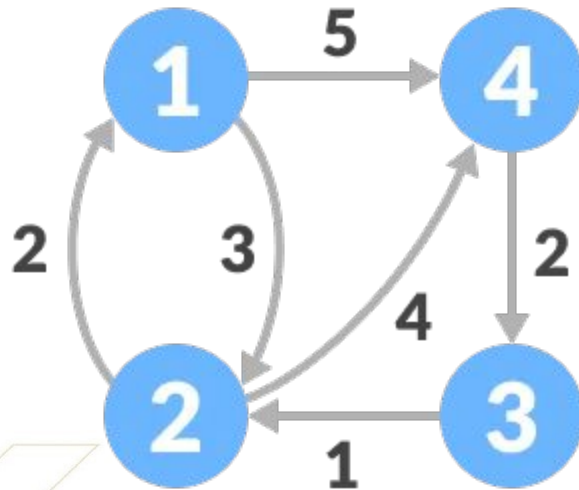
A^1

	1	2	3	4
1	0	3	INF	5
2	2	0	INF	4
3	INF	1	0	INF
4	INF	INF	2	0

A^2

	1	2	3
1	0	3	INF
2	2	0	INF
3	3	1	0
4	INF	INF	2

Floyd Warshall Algorithm



$$A^k[i, j] = \min (A^{k-1}[i, j], A^{k-1}[i, k] + A^{k-1}[k, j])$$

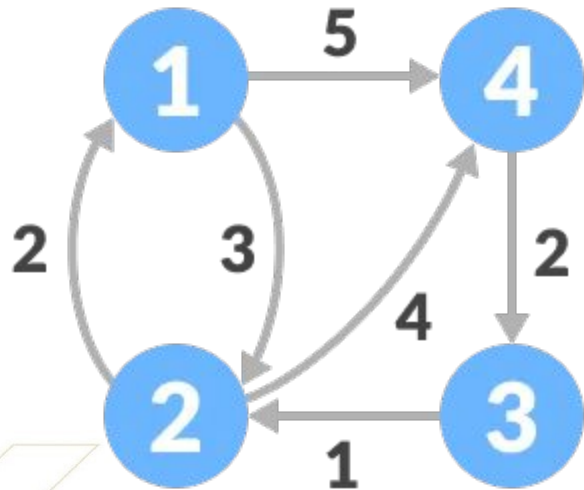
A^2

	1	2	3	4
1	0	3	INF	5
2	2	0	INF	4
3	3	1	0	5
4	INF	INF	2	0

A^3

	1	2	3	4
1	0	3	INF	5
2	2	0	INF	4
3	3	1	0	5
4	5	3	2	0

Floyd Warshall Algorithm



A^3

	1	2	3	4
1	0	3	INF	5
2	2	0	INF	4
3	3	1	0	5
4	5	3	2	0

A^4

	1	2	3	4
1	0	3	7	5
2	2	0	6	4
3	3	1	0	5
4	5	3	2	0

$$A^k[i, j] = \min (A^{k-1}[i, j], A^{k-1}[i, k] + A^{k-1}[k, j])$$

Floyd Warshall Algorithm

n = no of vertices

A = matrix of dimension n*n

for k = 1 to n

for i = 1 to n

for j = 1 to n

$A^k[i, j] = \min (A^{k-1}[i, j], A^{k-1}[i, k] + A^{k-1}[k,$

j])

return A

Time Complexity= $O(n^3)$

Space Complexity= $O(n^2)$

